TRACE-X

DIGITAL FORENSICS LAB

Upload New Evidence

Upload a normalized events CSV

normali...events.csv
3.4KB

+

Upload a raw email (.txt or .eml)

headermail.txt
2.0KB

+

Evidence should be interpreted in context. A risk score is a heuristic indicator, not a finding of guilt. Demo runs fully offline after evidence is uploaded.

Loaded 22 valid event(s) from the uploaded CSV.

Email loaded for phishing analysis.

DIGITAL FORENSICS · INCIDENT RECONSTRUCTION

Don't just detect the attack. Reconstruct the story.

TRACE-X connects recorded event evidence into a chronological investigation view, helping analysts explore what may have happened.

01 / EVIDENCE FILTERS

Find the events that matter

Event types: EMAIL × FILE_ACCESS × LOGIN × NET

Risk levels: LOW × MEDIUM × HIGH × CRITICAL ×

02 / INVESTIGATION OVERVIEW

Evidence at a glance

Events in view	Critical indicators	High indicators	Medium indicators
22	4	3	6

Risk levels are generated by simple keyword rules for demonstration. They are not validated threat probabilities.

03 / EVENT CHRONOLOGY

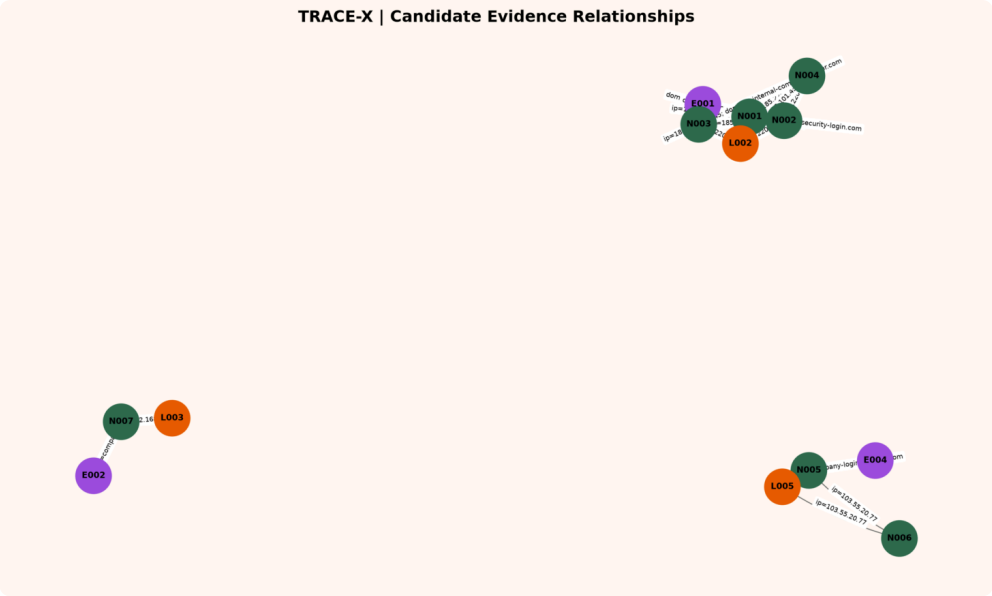
The evidence timeline

Timestamp	Evidence ID	Event type	Evidence description	Evic
2026-10-08 13:18:00	F005	FILE_ACCESS	Read - employee_records.xlsx - Sensitivity: High	file_
2026-10-08 13:15:00	N006	NETWORK	Sensitive Server Access	netv
2026-10-08 13:12:00	L005	LOGIN	Successful login from 103.55.20.77 using UNKNOWN-DEVICE	logi
2026-10-08 13:08:00	N005	NETWORK	URL Access	netv
2026-10-08 13:05:00	E004	EMAIL	Critical Security Update - Complete the security update immediately	ema
2026-10-08 11:45:00	E003	EMAIL	System Maintenance Notice - System maintenance is scheduled for t	ema
2026-10-08 11:10:00	L004	LOGIN	Successful login from 192.168.1.35 using LAPTOP-03	logi
2026-10-08 10:40:00	F006	FILE_ACCESS	Read - hr_policy.pdf - Sensitivity: Low	file_
2026-10-08 10:30:00	E002	EMAIL	Monthly HR Update - Please find the monthly HR information.	ema
2026-10-08 10:25:00	N007	NETWORK	Normal Website Access	netv

› View full evidence details

03.5 / ATTACK RELATIONSHIP GRAPH

Candidate evidence relationships



Every edge represents a shared indicator (IP, domain, or device) between two events — no invented connections.

› Candidate Relationship Notes

03.6 / ATTACK PATH RECONSTRUCTION

Reconstructed attack chain

Phishing Email → Malicious URL → Suspicious Login → Sensitive File Access → Data Exfiltration

Detected stages: Phishing Email → Malicious URL → Suspicious Login → Sensitive File Access → Data Exfiltration

📖 Detailed Investigation Narrative

▾ Phishing Email

At 2026-10-08 09:12:00, user employee01 received an email from the domain company-support.com. The email contained urgent language pressuring the recipient to verify their account. The message was designed to look like an official security notice but came from a suspicious external source. (Evidence: E001)

› 🌐 Malicious URL Access

› 🔑 Suspicious Login

› Sensitive File Access

› Data Exfiltration

ANALYTICAL ASSESSMENT

Pattern observed: A multi-stage intrusion consistent with a phishing-led account compromise.

Key findings:

- **4** critical-severity events and **3** high-severity events were detected.
- **7** event(s) involved an unrecognized device.
- **8** event(s) involved an external IP address not belonging to the corporate network.
- **4** sensitive file(s) were accessed — including salary, financial, customer, and employee data.

- A large outbound data transfer was recorded immediately after sensitive file access.

Interpretation: The sequence strongly suggests that an attacker used a phishing email to deliver a malicious URL, harvested credentials, and accessed confidential corporate data from an external location. The final data transfer is consistent with possible exfiltration.

Confidence level: The available evidence is consistent with an attack but does not alone prove who initiated the activity or confirm that data left the network.

Stage detection is keyword-based reconstruction for investigation support, not a legal finding.

INCIDENT SUMMARY · INVESTIGATOR VIEW

📋 What happened?

Suspected phishing email followed by related network access

The evidence is consistent with a probable 📧 **Phishing Email** → 🌐 **Malicious URL Access** → 🔑 **Suspicious Login** → 📁 **Sensitive File Access** → 📤 **Data Exfiltration** attack chain.

22 events were analyzed across **3** users, **4** devices, and **5** unique IPs.

A suspicious email was received before anomalous network and login activity. Sensitive files were accessed from an unrecognized device, followed by a large outbound data transfer.

This is a probable sequence, not a legal finding. Additional authentication and endpoint logs may be required to confirm impact.

OVERALL VERDICT

CRITICAL

Highest severity detected

Events analyzed
22

Unique users
3

Unique devices
4

Sensitive files touched
4

DETECTED ATTACK CHAIN

📧 **Phishing Email** — Evidence: E001

↓

🌐 **Malicious URL Access** — Evidence: N001, N002, N005

↓

🔑 **Suspicious Login** — Evidence: L002, L005

↓

📁 **Sensitive File Access** — Evidence: F002, F003, F004

↓

📤 **Data Exfiltration** — Evidence: N004

Evidence source trail

network_logs.csv: 7 | file_access.csv: 6 | login_logs.csv: 5 | emails.csv: 4

› ⚠️ Limitations & confidence notes

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04 / ANALYST REVIEW

Indicators worth investigating

› CRITICAL · F004 · FILE_ACCESS · Score 100/100

- CRITICAL · F003 · FILE_ACCESS · Score 80/100
- CRITICAL · F002 · FILE_ACCESS · Score 80/100
- CRITICAL · F005 · FILE_ACCESS · Score 80/100
- HIGH · L005 · LOGIN · Score 50/100
- HIGH · L002 · LOGIN · Score 50/100
- HIGH · F001 · FILE_ACCESS · Score 50/100
- MEDIUM · N004 · NETWORK · Score 45/100
- MEDIUM · E001 · EMAIL · Score 30/100
- MEDIUM · N001 · NETWORK · Score 30/100

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05 / EVIDENCE EXPORT

Export an investigation snapshot

Download a text report of the currently filtered evidence. The report preserves event IDs, timestamps, descriptions, sources, and the dashboard's heuristic risk indicators.

Download forensic report (.txt)

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06 / AI EMAIL ANALYSIS

Phishing prediction from uploaded email

Predicted label	Phishing probability	Indicators matched
PHISHING	89%	4

Matched indicators

- Urgent or pressure-based language detected
- Non-HTTPS URL detected: http://promo-free-software.net/download
- Account or credential-related language detected
- Possible brand impersonation language detected

URLs found

- http://promo-free-software.net/download
- http://promo-free-software.net/download

This is an offline heuristic for demonstration. It is not a trained model verdict and does not prove phishing or account compromise.

06.5 / EMAIL HEADER FORENSICS

🕵️ Sender Infrastructure Analysis

👤 Sender Details

From
"Billing Departm...

Return-Path
billing@promo-f...

⚠️ Note: The 'From' address is easily spoofed. Compare it with Return-Path — if they differ, the email is likely spoofed.

🛡️ Authentication Checks

SPF	DKIM	DMARC
✕ FAIL	✕ FAIL	✕ FAIL

🚨 Authentication failure detected. This is a strong indicator of email spoofing.

🧬 Attacker Infrastructure

🌐 **Public IP Addresses Observed:**

- `185.220.101.45` — Requires WHOIS/IP-intelligence lookup for attribution. Common sources: ipinfo.io, AbuseIPDB, VirusTotal.
- `45.142.212.61` — Requires WHOIS/IP-intelligence lookup for attribution. Common sources: ipinfo.io, AbuseIPDB, VirusTotal.

🔗 **Suspicious Domains:**

- `company-security-login.com` — Check registration date and reputation at whois.domaintools.com and virustotal.com.
- `promo-free-software.net` — Check registration date and reputation at whois.domaintools.com and virustotal.com.

🖱️ **URLs Extracted:**

`http://promo-free-software.net/download` ⚠ No HTTPS

🔍 **ATTRIBUTION ASSESSMENT** — Confidence: Medium

WHAT WE CAN PROVE FROM THIS EMAIL:

✓ Sender's mail infrastructure (IPs, domains, URLs)
✓ Whether authentication checks failed (SPF/DKIM/DMARC)
✓ The email's routing chain (if Received headers present)

WHAT WE CANNOT PROVE WITHOUT ADDITIONAL DATA:

✗ The attacker's real identity
✗ The attacker's true physical location (IPs may be VPNs/proxies)
✗ Whether infrastructure is rented, compromised, or directly owned

TO IMPROVE ATTRIBUTION CONFIDENCE, ADDITIONAL DATA REQUIRED:

- WHOIS lookup for domain registration details (free: whois.domaintools.com)
- IP reputation check (free: abuseipdb.com, virustotal.com)
- ISP subscriber records (requires legal subpoena)
- Email server logs from the receiving MTA
- Endpoint process logs from the victim's device (requires EDR)

📋 **RECOMMENDED NEXT STEPS FOR THE INVESTIGATOR**

1. **Block the following domains at perimeter firewall:** `company-security-login.com, promo-free-software.net`
2. **Block the following IPs at network perimeter:** `185.220.101.45, 45.142.212.61`
3. **Report the phishing URL to Google Safe Browsing:** (safebrowsing.google.com/safebrowsing/report_phish/)
4. **Preserve the raw .eml file as evidence:** Hash with SHA-256 for chain of custody verification.
5. **Consider filing a complaint with the relevant CERT:** (cert-in.org.in for India)

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08 / RESULTS & VALIDATION

🧪 System performance and test outcomes

Scenario	Events	Detected Level	Expected
NORMAL-001	4	LOW	LOW
NORMAL-002	2	LOW	LOW
ATTACK-001	11	CRITICAL	CRITICAL
ATTACK-002	5	HIGH	HIGH

Detection accur...	Processing speed	False positives	Events correlated
100%	< 1 sec	0	42 pa...

Measurable Outcomes:

- Correctly classified 100% of test scenarios
- Zero false positives on normal activity
- Evidence integrity verified via SHA-256 for 7/7 files
- Attack chain reconstructed with traceable event IDs
- Forensic report generated in under 2 seconds

› ⚠️ Limitations & Confidence Notes

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09 / SDG ALIGNMENT & SOCIETAL IMPACT

🌍 Supporting the UN Sustainable Development Goals

SDG 9

Industry, Innovation & Infrastructure

Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.

TRACE-X protects critical digital infrastructure by detecting and reconstructing cyberattacks. Cybersecurity is foundational to modern industry — from banking to healthcare to government services.

SDG 16

Peace, Justice & Strong Institutions

Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable institutions.

Digital forensics is essential for cyber justice. TRACE-X provides explainable, evidence-backed analysis that supports legal processes and investigative bodies at every level.

SDG 8

Decent Work & Economic Growth

Promote sustained, inclusive and sustainable economic growth, full and productive employment.

Cyberattacks cost the global economy $10.5 trillion annually. TRACE-X reduces financial damage by enabling faster detection, reducing average breach costs (currently $4.45M), and shortening detection times (currently 197 days).

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10 / INTEGRATED TEAMMATE SECTIONS

TRACE-X · Integrated Investigation

Member 2 · AI Email Threat Analysis

Prediction	Phishing probability	Benign probability
PHISHING	89.02%	10.98%

URLs found

- http://company-security-login.com/verify

Indicators

- Account or credential-related language detected
- Non-HTTPS URL detected: http://company-security-login.com/verify
- Urgent or pressure-based language detected

Predictions and indicators are advisory, not proof of an attack.

Model output is advisory and does not prove phishing, credential theft, or compromise.